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Perspective

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Surfaces, interiors and evolution of solar system moons

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This Perspective reviews what we have learnt about the major moons of the solar system over the past 30 years. Major discoveries include subsurface oceans in at least five of these moons, geological activity on at least three (Io, Triton and Enceladus), and an active hydrological cycle (involving hydrocarbons) on Titan. Geological activity, either current or past, is in most cases driven by tidal heating; thus, the orbital evolution of these moons is intimately connected to their geological evolution in a way that is not true of planets in this solar system. The rate of evolution depends on tidal dissipation in the parent planet, which is very poorly understood but may in some cases be larger than previously thought. How the satellites formed can in principle be deduced based on observations such as the extent of volatile depletion or degree of differentiation, but unique answers are elusive. In the early 2030s, a flotilla of missions including Europa Clipper, JUICE, Dragonfly, Tianwen-4 and MMX will solve some of the existing major puzzles identified here, and introduce new ones.

1. Introduction

The moons of the solar system present a dazzling array of diversity, from hyperactive Io to cloud-shrouded Titan to lumpy Miranda (figure 1). The aim of this Perspective is to summarize what we have learnt about the surfaces and interiors of these bodies over the last 30 years, and to investigate some of the possible origins of the diversity we see.

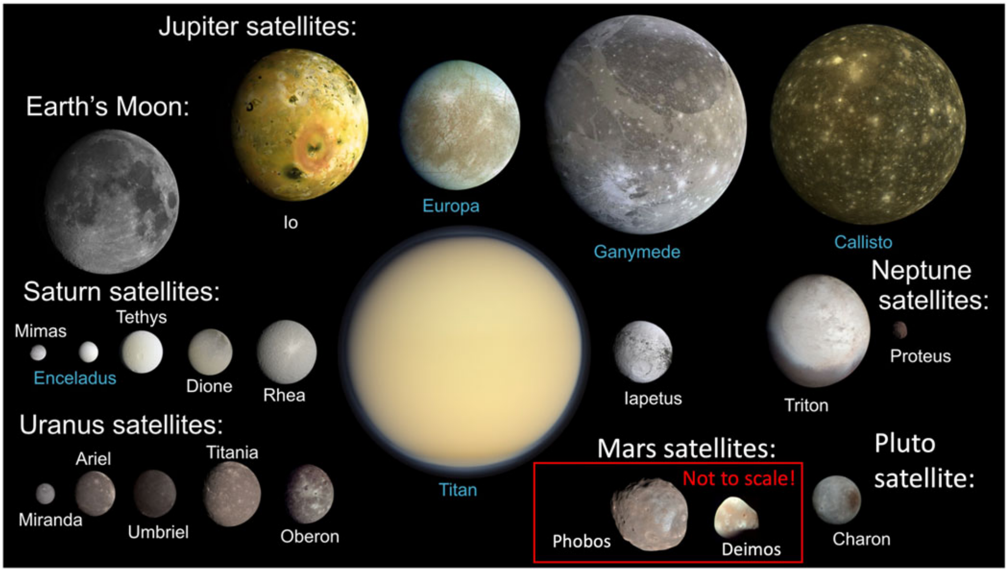

Figure 1. Major moons of the solar system, modified from Nimmo & Pappalardo [1]. Everything is to scale except the small Martian moons Phobos and Deimos (they are 22 km and 13 km across, compared to 3500 km for our Moon). Blue labels indicate moons that likely possess subsurface oceans.

Moons are important for several reasons. They are complex, because their thermal evolution is frequently coupled to their orbital evolution [2], which is not the case for planets (at least in this solar system). Their existence and characteristics can tell us about processes (such as impacts) and conditions (such as temperatures) operating as the planets formed [3]. And because several moons are thought to possess liquid water beneath the surface, they provide examples of where potentially habitable environments (and perhaps life itself) could arise [4].

To make these topics more accessible, below I will generally use qualitative, rather than quantitative, arguments. The latter may be found in the references provided in the text. I cite selected papers as examples rather than attempting to provide an exhaustive survey of the literature (which is large). There are also excellent books covering many of the topics examined below in much more detail. Both the University of Arizona Press and Cambridge University Press have issued books on individual moons such as Europa, Enceladus and Titan, as well as books that describe particular sub-topics, such as *Planetary Surfaces* [5]. Another useful resource is the most recent US Planetary Sciences Decadal Survey [6].

Which moons are of interest? A moon is a body that orbits a planet, a dwarf planet (like Pluto) or a smaller body such as an asteroid. In this review, I will generally focus on the larger moons and pay less attention to the small moons orbiting asteroids and the small irregular satellites in the outer solar system. A convenient definition of 'small' is the size below which the yield stress of the material making up the body is larger than its central pressure [5]. A typical yield stress of 10 MPa yields a cutoff radius of about 100 km. All the moons shown in figure 1 are larger than this value, except the two moons of Mars, Phobos and Deimos (11 km and 6 km radius, respectively). Some Kuiper Belt Objects have sizeable moons [7] but (apart from Charon) these are not well characterized and will mostly be neglected. In the discussion below, I will arbitrarily refer to the largest seven bodies—Io, Europa, Ganymede, Callisto, Titan, Triton and our own Moon—as 'large' moons and the others in figure 1 (except Phobos and Deimos) as 'medium-sized' moons.

The rest of this article is arranged as follows. I start by discussing the interiors of these bodies: what we have learnt about their internal structures, and what this tells us. I will then go on to

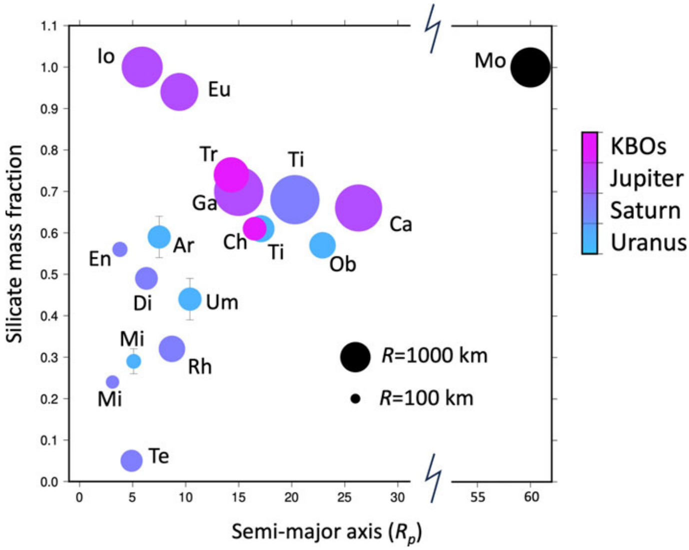

Figure 2. Rock mass fraction vs. semi-major axis for solar system moons. Semi-major axis is given in terms of primary radii (R_p) and rock mass fraction is calculated based on measured bulk density assuming rock and ice densities of 3.5 g cc^{-1} and 0.95 g cc^{-1}, respectively. Abbreviations are as follows: Ar(iel), Ca(llisto), Ch(aron), Di(one), En(celadus), Eu(ropa), Ga(nymede), Mi(randa) [Uranus], Mi(mas) [Saturn], Ob(eron), Pl(uto), Rh(ea), Te(thys), Ti(tan) [Saturn], Ti(tania) [Uranus], Tr(iton), Um(briel), Mo(on). Error bars for Uranian satellites reflect uncertain masses.

discuss their surface features. Finally, I will present a briefer discussion about how these bodies may have formed and evolved over time.

2. Interiors

The study of planetary interiors includes determining their composition, structure and dynamics. Here composition refers to the bulk materials from which they are made, while structure means how those materials are distributed and dynamics focuses on material motion. These three sets of characteristics can then be used to make inferences about the origin, evolution and present-day behaviour of the bodies.

(a) Composition

For most moons, we have two main sources of direct information about their composition. First, remote sensing, and particularly reflectance spectroscopy, can be used to deduce the mineralogy of the surface [8]. Icy satellites are so-called because the dominant spectral component is water ice, though various contaminants (such as NH_3 [9], CH_4 [10], NaCl [11] or condensed volatiles such as CO_2 [12]) may also be present. The presence or the absence of such species may help determine what conditions applied during satellite formation (§4a). Some non-ice species, and NH_3 in particular [13], can substantially reduce the freezing temperature of subsurface oceans (figure 3); sulfur has the same anti-freeze effect for iron cores [14].

In the case of moons with subsurface oceans, non-ice surface species such as NaCl may give clues to ocean composition (e.g. [15]). However, surface composition can also be affected by exogenic processes such as bombardment by micrometeorites or energetic particles (e.g. [16]). Enceladus is unusual in that the composition of pristine ice-dominated material can be measured directly via *in situ* techniques during spacecraft flybys through the south polar plume (e.g. [17]).

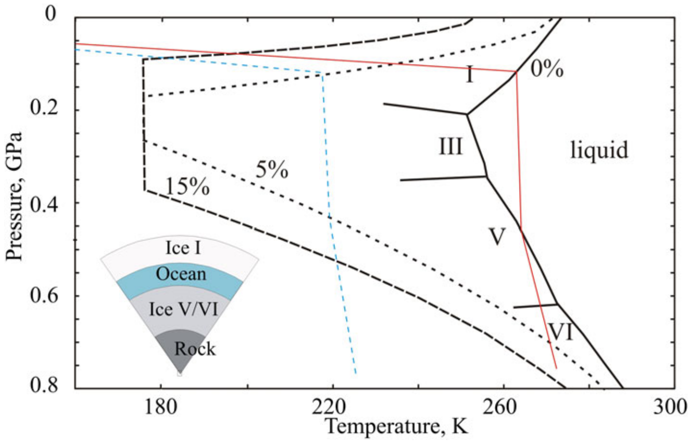

Figure 3. Ice melting behaviour and phases, after Hussmann *et al.* [26]. Solid lines represent pure H_2O phase diagram, with different phases marked with Roman numerals. Dashed lines represent melting curve of H_2O plus 5% or 15% ammonia. Thin solid red and blue dashed lines show hypothetical thermal profiles, illustrating how a liquid layer may develop between low- and high-pressure solid phases. Note that in reality the oceans would be adiabatic and not isothermal. The inset shows the generic structure of a large icy satellite such as Ganymede.

Measurements of isotopic ratios, useful for assessing provenance, typically require *in situ* sampling. However, in some cases remote sensing can be used. Two important examples are D/H on the Saturnian moons [18] and (remarkably) sulfur and chlorine isotope ratios on Io [19].

The second way of assessing a moon's bulk composition is to measure its bulk density. Cosmochemical arguments suggest that most moons should be some mixture of rock (plus metal) and water ice, so a denser moon probably possesses more rock (density $\approx 3\,\mathrm{g\,cc^{-1}}$) and less ice (density $\approx 1\,\mathrm{g\,cc^{-1}}$). A couple of caveats are in order. First, in the outer solar system other components (such as organic materials found in comets [20]) may be present in significant quantities; if three major components are present, no conclusions can be drawn about the composition of the object from density measurements alone. Second, at low pressures both ice and rock can be porous and therefore have a density lower than the canonical value. A good example is Enceladus, which appears to have a low-density rock core [21] (either because of porosity or because it has been chemically altered). Conversely, high-pressure ice phases (see below) will complicate interpretation of rock–ice ratios.

Bearing these caveats in mind, the bulk density can be used to crudely infer the rock–ice ratio of solar system moons. Figure 2 shows the remarkable variability that results. Some moons, like Tethys, are balls of almost pure ice. Others, like Io or the Moon, are entirely rock plus metal. Many large moons have an apparent rock : ice ratio of roughly 2 : 1, which also appears characteristic of Kuiper Belt Objects [22]. This raises a puzzle, since condensation of solar material is expected to yield a rock : ice ratio more like 1 : 1 to 1 : 3 [23].

Smaller moons tend to have lower densities on average, but the scatter is large. Part of the explanation might be due to porosity, but the very low inferred rock mass fraction of Tethys strongly suggests that it is a fragment from some impact involving a larger moon [24]. The Galilean satellites show a striking progression of decreasing rock mass fraction with increasing distance, for reasons that are debated (§4). This progression is not, however, seen for either the Uranian or the Saturnian satellites.

(b) Structure

The three most important questions regarding the internal structures of moons are: (i) are they differentiated?, (ii) do they have liquid layers?, (iii) if so, how close to the surface does the liquid get? Differentiation simply means that the dense components have sunk to the centre of the body; an undifferentiated moon would be a homogeneous mixture of rock and ice, for example. A liquid layer could be a water ocean beneath an ice shell, a magma ocean or a liquid iron core. For an ocean, the proximity to the surface (i.e. the shell thickness) is important because it influences the likelihood of exchange between the ocean and surface. This matters both for understanding the origin of surface features, but also because it affects our ability to measure ocean characteristics like composition with remote sensing.

(i) Expectations

Based on §2a, we would expect most moons to consist primarily of rock (plus metal) and ice. Porosity is important for small (asteroid-sized) bodies but is probably of minor importance for the medium-sized or large moons. As discussed below, if temperatures are sufficiently high the moons will separate into rock below and ice above. Even higher temperatures might cause any metal present in the 'rock' to melt and form a discrete core.

For differentiated bodies, the phase diagram of ice (e.g. [25]) has some crucial properties. The low pressure phase of ice (ice I) has a melting temperature that decreases with increasing pressure. As a consequence, ice has a lower density than liquid water, naturally resulting in an insulating lid sitting atop any subsurface ocean. Higher-pressure phases of ice, however, are denser than water so—if present—they would sit beneath the ocean. The pressures required (greater than 0.2 GPa) imply that such high-pressure ice phases are only likely on the large moons. In such cases, the red line in figure 3 illustrates that the ice shell thickness then also determines the ocean thickness, because the temperature in the ocean itself is presumed to follow an adiabat.

Another important consequence of ice I's unusual density is that a refreezing ocean will expand. As a consequence, such refreezing will lead to surface extension [27] and possible ocean pressurization and eruptions [28] (§3c). However, for the large moons this effect will be largely counteracted by the opposite effect for higher-pressure ice phases [29].

Finally, some contaminants, and ammonia in particular, can significantly reduce the melting temperature of water ice. Figure 3 shows that in the case of ice I, addition of 15 wt% NH_3 can depress the melting temperature by almost 100 K.

Phase diagrams are somewhat less complicated for magmatic and iron-dominated species. In both cases, melting temperatures increase with increasing pressure, and the liquid (melt) is usually less dense than the solid. The melting temperature is lowered by addition of contaminants, such as water (in the case of magmas) or sulfur (in the case of iron).

(ii) Differentiation

The reason that differentiation is important is because it tells us about the thermal history of a body. The precursor bodies from which moons are made most likely started out undifferentiated. For differentiation to happen, heat needs to be supplied; differentiation is very rapid once melting starts. There are four main relevant sources of heat.

(i) Decay of ^{26}Al is a strong heat source, but only operates during the first 3–4 Myr of solar system history.
(ii) As objects accrete, they gain thermal energy. This process only applies during the accretion process itself; the resulting mean temperature increase goes roughly as $3GM/5RC_p$, as long as accretion is fast enough to avoid radiating all the heat away (here M,R and C_p are the body mass, radius and specific heat capacity). For the large moons, this temperature increase is comparable to that required to melt ice, but not rock or metal.
(iii) Radioactive decay of long-lived elements (K,U,Th) causes temperatures to increase by an amount that depends on how rapidly heat is removed (e.g. by conduction or convection). Larger and more rock-rich bodies experience bigger temperature increases. For the large moons, the temperature changes can be sufficient to melt rock or metal (e.g. [30]).

Table 1. Inferred moments of inertia for moons. Sources are given in Notes. UHA, unverified hydrostatic assumption (see text). NHR, non-hydrostatic effects removed (see text).

object	MoI	notes/refs	object	MoI	notes/refs
Moon	0.394	[32]	Titan	≈ 0.341	[33]; NHR
Io	0.37824 ± 0.00022	[31]	Enceladus	≈ 0.331	[21]; NHR
Callisto	0.3549 ± 0.0042	[31]; UHA	Dione	≈ 0.33	[34]; NHR
Europa	0.3547 ± 0.0024	[35]; UHA	Ganymede	0.3115 ± 0.0028	[31]; UHA

(iv) Tidal heating is discussed in more detail below (§2c(i)). In some cases, tidal heating is sufficient to generate global melting (Io), while in other cases it is negligible. An important aspect of tidal heating is that, unlike other heat sources, it can be non-monotonic.

To determine whether an object is differentiated, we need to determine its moment of inertia (MoI). Various methods are available to do so, most of which rely on measuring the body's gravity, topography, and/or axial tilt (measurements that can be made without landing a spacecraft on the surface). A MoI of 0.4 indicates a completely homogeneous object; a value less than 0.4 indicates more mass is concentrated towards the centre (i.e. some degree of differentiation has occurred). However, even for a two-layer body, measurement of the bulk density and MoI is insufficient to completely determine the interior structure: an additional assumption (e.g. density of the outer layer) has to be made. More complicated models are correspondingly more non-unique. A useful summary of these issues as applied to the Galilean satellites is in [31].

Table 1 lists the moons where a reliable MoI value has been determined. Typically, these measurements are carried out by measuring the body's long-wavelength (degree-2) gravity and relying on an assumption that the moon is responding as a fluid (so-called 'hydrostatic equilibrium'). Sometimes, this assumption can be verified by other measurements, and sometimes non-hydrostatic effects can be quantified and removed, but in other cases (especially Callisto; [36]) neither is possible and thus the MoI value is provisional (and the formal errors quoted should not be taken seriously).

At one end of the spectrum, our Moon appears almost homogeneous (MoI $\approx$ 0.4). This is because the Moon is mostly silicates, with only a small ($\approx$ 400 km radius) iron core at the centre. Io also consists of a core and silicate mantle, but its core is bigger and so its MoI is lower. At the other end of the spectrum, Ganymede has the lowest known MoI of any solid body. Because Ganymede possesses a dynamo (see below), it must consist of an at least partly liquid iron core, overlain by a silicate mantle and then a thick ice shell. Ganymede also appears to have localized gravity anomalies, of uncertain origin [37]. Callisto, with almost the same bulk density as Ganymede, apparently has a markedly higher MoI (based on an unverified assumption of hydrostatic equilibrium). If true, this means that Callisto is not fully differentiated, which means it must have accreted slowly, a very important constraint (§4a). Europa's MoI is similar to that of Callisto, but because its bulk density is much higher (figure 2) the interpretation is that Europa only possesses a thin ($\approx$ 100 km) ice–water layer above a rocky interior. Enceladus appears to be fully differentiated but with a silicate interior that has a low density ($\approx 2500\,\mathrm{kg\,m^{-3}}$), indicating either porosity or chemical alteration [21]. Titan also appears to have a differentiated structure and low density, perhaps chemically altered, silicates [38]. Finally, the apparent MoI of Dione is consistent with complete differentiation, but the uncertainties are sufficiently large that any inferences should be treated with considerable caution.

(iii) Liquid layers

Liquid layers are important for two main reasons. First, the existence of such layers implies an energy source sufficient to keep parts of the moon above the melting temperature. As with differentiation, this energy source could result from stored primordial heat (due to accretion or ^{26}Al decay), long-lived radioactive decay or tidal heating. Second, liquid water oceans are of particular interest because they represent potentially habitable environments, where life could conceivably have developed [4].

As reviewed in [1], there are a variety of ways of detecting liquid layers. In brief, the four most common are: induction, tidal response, librations and obliquity. In the induction technique, a conductive layer (a salty water or magma ocean) is detected because it generates an induced magnetic field when subjected to a time-varying background field. A shell that is decoupled from the interior by an ocean exhibits a large tidal distortion; this can be detected via either gravity or altimetry measurements (e.g. [39,40]). This decoupling also allows the tidally induced backwards-and-forwards motion of the shell (libration) to be larger than expected for a solid object. Last, the obliquity (tilt) of a satellite can be used to deduce whether a decoupled layer exists. A satellite exhibiting compensated topography (as at Enceladus) might be doing so because of shell thickness variations above a subsurface ocean, but other possibilities (e.g. lateral density contrasts) also exist.

Thanks to the induction technique, oceans are fairly certain to exist beneath the surfaces of Europa, Ganymede and Callisto [41,42]. Titan also probably has a subsurface ocean, though here the inference is based on obliquity [43] and tidal response [33] measurements. Enceladus is known to have a global ocean because of libration measurements [44]; plume samples analysed by *Cassini* indicate the ocean is salty [17].

For other icy bodies, we have to rely on theory. Radiogenic heating alone would be sufficient to maintain an ocean inside a body the size of Triton [45] as long as the ice shell is not convecting (see §2c(ii)). For smaller bodies, such as Charon or the Uranian satellites, present-day oceans would in addition require the presence of either a porous, low conductivity shell, some kind of anti-freeze, and/or tidal heating (e.g. [46]).

Based on its induction response, Io has a partly molten interior [47], but its tidal response rules out a shallow magma ocean [48]. The Moon did have such a magma ocean shortly after it formed, but this is thought to have crystallized by a few hundred Myr after solar system formation.

The Moon also has a liquid iron core, which was detected primarily by measurements of its rotation state [32]. Ganymede too must have such a core, because this is the only known way of generating its internal dynamo (see §2c(iii)). Large moons like Io or Titan that lack dynamos might still have iron cores that are liquid (but not convecting), but there is currently no observational evidence for their state.

(iv) Shell and elastic thickness

For ocean-bearing moons, the depth to the ocean, i.e. the shell thickness, is a key parameter because it is presumed to control the extent to which exchange of materials between the surface and ocean takes place, as well as affecting the morphology of surface features. Ideally it is determined via gravity and topography measurements and an assumption of isostasy, as at Enceladus [21]; an alternative is to use impact craters as natural probes (e.g. [49]).

More indirect techniques provide the heat flux at the time of deformation, which can then provide an estimate of the shell thickness. This is most easily accomplished by identifying a feature (e.g. a mountain on Io) causing elastic deformation of the lithosphere. The width of the deformation then allows the so-called elastic thickness T_e to be determined [50]. Since the base of the elastic layer is defined by an isotherm, roughly 60% of the melting temperature, determination of T_e allows the temperature gradient and thus the heat flux to be derived. In the case of an ocean world, the temperature gradient obtained can also be extrapolated down to the melting temperature to determine the shell thickness. This approach works if the shell is conductive; if it is convective only a lower bound on the total shell thickness can be obtained. The approach has been applied to many moons, including Europa [51], Enceladus [52], Tethys [53], Ganymede [54], Dione [55], Titania [56], Ariel [57] and Charon [58]. A similar technique deriving paleo-heat fluxes from the relaxation of impact craters has also been widely used (see §2c(ii)). For the Moon, the

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elastic thickness can also be determined by combining gravity and topography measurements (e.g. [59]), but this approach is not possible for other moons because the gravity data are not currently good enough.

Of the ocean-bearing moons (figure 1), the one with the best-known shell thickness is Enceladus. Here the mean shell thickness is about 20–25 km, with thicknesses as low as 6 km at the South Pole [60]. This is the region which is geologically active and spewing water vapour and ice grains into space, presumably supplied from the ocean via throughgoing fractures.

At Europa, the situation is less satisfactory, with shell thickness estimates ranging from a few km to a few tens of km thick. Probably, the best current values come from impact crater studies, which suggest thicknesses of around 20 km [49]. Io has a rigid lid above its molten or partially molten interior, but its thickness—of order 30 km [61]—is uncertain . For Ganymede, Callisto and Titan the inferred thickness of the shell is based primarily on our knowledge of the ice phase diagram (see §2b(i)), which ends up with an ice I thickness of order 100 km.

Finally, our own Moon does not have a subsurface magma ocean now, but the anorthositic highland crust probably did originally float on such an ocean, and has a thickness of about 40 km based on gravity, topography and seismic measurements [62].

(c) Dynamics

Dynamics, that is, motions in the interiors of the moons, require an energy source, usually (but not always) thermal. The main sources of thermal energy are radioactive decay and tidal heating. As a dynamic process itself, tidal heating will be discussed first, followed by the dynamics of solid and liquid layers. Tidal heating is intimately tied to a moon’s long-term orbital evolution (see below). The essential textbook dealing with orbital and tidal dynamics is *Solar System Dynamics* [63].

(i) Tidal heating

All the moons considered here rotate synchronously, so they always present the same face towards their primary. Because the gravitational potential of the primary varies spatially, these moons are elongated in the direction of the primary, resulting in a so-called tidal bulge. Furthermore, because moons’ orbits are usually not quite circular, the distance to the primary changes with time, and so too does the apparent position of the primary as seen from a fixed point on the moon’s surface. As a consequence of these two effects, the tidal bulge raised on the moon by the planet moves backwards and forwards relative to the moon’s surface. The amplitude of this time-variable (‘diurnal’) tide depends on how rigid the moon is. Any friction in the system converts some of this deformation energy to thermal energy, otherwise known as tidal heating. Tidal heating can be a dominant process: Io is the most volcanically active body in the solar system because of it, and it is the only viable explanation for the survival of Enceladus’s subsurface ocean [64].

The rate of heat production $\dot{E}$ due to tidal heating is given by [63]

$$\dot{E} = \frac{3}{2}\frac{n^5 R^5}{G}\frac{k_2}{Q}(7e^2 + \theta^2). \tag{2.1}$$

Here n, R, e and θ are the mean motion, radius, eccentricity and obliquity of the satellite, respectively. The quantity k_2 is its diurnal tidal Love number, a measure of the amplitude of the tidal response, and Q is its dissipation factor, which determines the rate at which mechanical energy is dissipated as heat (a low Q indicates high dissipation and a large phase lag between the forcing potential and the response). At present, the only satellites for which we have a measurement of k_2/Q are the Moon [32], Io [65] and Titan [66].

The orbit of an isolated satellite experiencing tidal heating will become more circular, thus shutting tidal heat off. Thus, Triton likely experienced a transient episode of extreme tidal heating as its original orbit circularized [67]. Persistent tidal heating requires a so-called orbital resonance (such as that between Enceladus and Dione), in which two or more moons excite each others’ eccentricities, counteracting the eccentricity-damping effect. In this case, the ultimate source of energy is the rotational kinetic energy of the planet, which is typically a very large reservoir. As a consequence, tidal heating at places like Io and Enceladus may have persisted for billions of years. Since long-term heating requires a satellite to be in a resonance, the orbital evolution of satellites may have controlled their internal dynamics and surface tectonics. Thus, for instance, the Uranian satellites are not in any orbital resonances now, but may have experienced tidal heating events in the past [57], when their orbital configurations were different [68].

Another consequence of eccentricity damping is that an isolated satellite with an unexpectedly high eccentricity must either not be very dissipative, or have had its eccentricity excited recently. Because dissipation depends on the internal structure of the body (e.g. warm bodies are more dissipative—higher k_2/Q—than cold bodies), such analyses can provide a useful constraint. A good example is Titan, which has a high eccentricity (0.028) but is also apparently quite dissipative [66]. Recent arguments [69] suggest that Titan's eccentricity is the result of a recent (few hundred Myr), violent orbital disturbance within the Saturn system, which may also explain Saturn's spectacular rings.

Direct measurements of tidal heating are scarce. Io's tides yield a prodigious heat flux of about $2.5\,\mathrm{W\,m^{-2}}$ [70] (the Earth's mean heat flux is $0.08\,\mathrm{W\,m^{-2}}$). The Moon is much colder than Io and therefore much less dissipative; consequently, tidal heating does not significantly affect its present-day thermal state. Enceladus's mean tidal heat production is somewhat uncertain but is probably around $0.04\,\mathrm{W\,m^{-2}}$ [64]. For comparison, primitive bodies produce radioactive heat at present at a rate of about $10^{-8}\,\mathrm{W\,m^{-3}}$ [50]. A rocky body comparable to Enceladus (250 km) or Io (2000 km) would produce radiogenic heat fluxes of about 1 and $7\,\mathrm{mW\,m^{-2}}$; these values would be reduced by the presence of ice.

Although eccentricity tides are the most common source of tidal heating, there are also obliquity tides. In this case, the moon is tilted and the tidal bulge moves north and south over the course of an orbit. Obliquity tides otherwise operate in a similar manner to eccentricity tides; they cause the orbital inclination to damp [71] and are thus self-limiting unless a resonance is present. One place where obliquity tides may be dominant at present is Triton, by virtue of its circular (but inclined) orbit [45].

(ii) Dynamics of solid layers

One very important process potentially affecting solid layers (rocky or icy mantles) is solid-state convection. A layer with a sufficiently low viscosity will convect, moving heat outwards more rapidly than if the shell is motionless and conductive. Convection can therefore inhibit the development of subsurface oceans, by cooling the body too rapidly. It may also be able to produce surface features (see below).

Whether convection occurs depends on the Rayleigh number Ra, which is defined as

$$Ra = \frac{\rho g \alpha \Delta T d^3}{\kappa \eta}. \tag{2.2}$$

Here ρ is the density, g the gravity, α the thermal expansivity, ΔT the temperature contrast across the layer, d the layer thickness, κ the thermal diffusivity and η the viscosity. For a constant-viscosity fluid, the Rayleigh number needs to exceed about 10^3 for convection to occur. For a variable-viscosity fluid (like ice or rock), η is taken to be the viscosity at the bottom of the layer and the critical Rayleigh number is larger, typically 10^6 or 10^7 (e.g. [72]).

The most uncertain factor in these calculations is almost always the viscosity. For ice and rock near their melting temperatures, ballpark values are approximately 10^{14} Pa s and approximately 10^{21} Pa s, but these are strongly temperature-dependent and also depend on the relevant grain-size. For ice shells floating on oceans, a pure water ocean will be at about 270 K and so the ice shell viscosity will be close to 10^{14} Pa s. However, if the ice shell contains anti-freeze, the ocean temperature will be lower, the viscosity higher and the Rayleigh number lower. Thus, adding anti-freeze to an ocean is an effective way of shutting off ice shell convection.

There is currently no direct evidence for convection in moon interiors. Models suggest that our Moon's mantle may be sluggishly convecting at the present day [73], and convection in the thick ice I shells of Titan, Callisto and Ganymede is likely (but not certain) [74]. Europa's ice shell could conceivably be convective. On the other hand, Enceladus's ice shell cannot be convective—if it was, the observed shell thickness variations would be rapidly smoothed out (see below).

Though convection is usually driven by thermal density contrasts, convection can also be driven by compositional contrasts. One possible example is the presence of diapirs at Triton (see §3b) which might be due to rising buoyant material [75].

Apart from its effect on the heat budget, convection can have other consequences. For instance, rising and sinking convective blobs will alter the body's moments of inertia. This in turn can lead to reorientation of the solid body ('true polar wander') and generate stresses which can be recorded on the surface (see §3b). One place where this likely happened is Miranda [76].

Convection is not the only way of getting heat out. Io is partially molten on the inside, and compositional buoyancy differences drive this melt upwards to produce volcanism. It is this erupting material that transports Io's prodigious heat budget to the surface, a so-called 'heat-pipe' mechanism [77], which may also have operated on the early Moon [78]. We do not expect similar mechanisms to exist on icy bodies for the simple reason that water is more dense than ice and is thus much less likely to erupt (see §3c).

Another important process is lateral flow of the crust/shell. If there are lateral thickness variations in the crust, there will also be pressure gradients that drive lateral flow of the crust. This flow will act to reduce the amplitude of the initial variations. The rate of flow depends on the viscosity and thickness of the flowing material. In a conductive crust, the warmest material at the base of the crust will flow most rapidly, so that flow is confined to a relatively thin channel. In a convective crust, flow will not be so confined and so relaxation is correspondingly more rapid.

Topographic relaxation caused by lateral flow is most rapid at short wavelengths.[1] Because ice has a low intrinsic viscosity, lateral flow is effective at wiping out shell thickness variations except for very thin shells or at very long wavelengths [79]. The pronounced shell thickness variations inferred at Enceladus [60] rule out ice shell convection, because the thickness variations would be erased very rapidly. The limited long-wavelength topography seen at Europa may be a consequence of lateral shell flow [80].

Lateral flow can also be driven by surface topography rather than crustal thickness variations. For instance, impact craters produce lateral pressure gradients. Since these gradients only extend to depths comparable to the crater diameter, larger craters are more likely to undergo relaxation, as they drive flow at greater depths where material will be warmer and thus flow more rapidly. Apparently relaxed craters or basins are found on many bodies, including Enceladus [81], Dione and Tethys [82], Ganymede [83] and our own Moon [84]. These relaxed features are useful because they allow us to determine what the likely heat flux was.

(iii) Dynamics of liquid layers

Although liquid core dynamics have been studied for decades, only in the last few years has there been significant attention paid to subsurface ocean dynamics. As yet, there are few if any observational constraints allowing different hypotheses regarding the latter to be distinguished. Furthermore, there is not yet universal agreement on the best modelling approach; as with liquid cores [85], the actual parameter space occupied by these oceans is not accessible with present-day computer power. An excellent overview of the state of the art may be found in [86].

Because both water and liquid iron have low viscosities, their Rayleigh numbers (equation (2.2)) tend to be very large. However, this does not necessarily mean that they are convecting, because there is also a minimum heat flux required to drive convection, the so-called adiabatic value[2] [87]. For a liquid iron core like that of Ganymede, the adiabatic heat flux is of order

[1] Note that short-wavelength topography can also be supported elastically, without requiring shell thickness variations.

[2] The adiabatic heat flux is given by $k\alpha g T/C_p$ where k is thermal conductivity, T is the temperature and C_p the specific heat capacity.

5 mW m^{-2}, while for a Europa-like ocean it is of order 0.02 mW m^{-2}. Comparison with the heat flux estimates above show that core convection is not guaranteed, whereas ocean convection is (unless impeded by salinity gradients; e.g. [88]).

Whether or not liquid cores convect is important because convection is a requirement to drive a dynamo and generate an internal magnetic field [87]. Ganymede's core must therefore be convecting; but the absence of dynamos at Io, Europa, etc., might simply be because the heat extracted from the core is sub-adiabatic, and does not require these bodies to have a solid core [89]. Core convection can also be aided by compositional effects—for instance, iron crystals may solidify at the top of the core and then sink due to their excess density [90].

The circulation patterns of both cores and oceans are strongly influenced by the rotation of the moons. Typically, this is rapid enough that circulation patterns form long cylinders ('Taylor columns') parallel to the rotation axis. This geometry means that regions inside and outside the so-called tangent cylinder may deliver quite different heat fluxes to the areas above (e.g. [91]). Thus, in principle ocean circulation patterns could be inferred by looking at variations in ice shell thickness—as long as lateral flow (§2c,(ii)) is not dominant. Conversely, if shell thickness variations exist, then they are expected to drive long-wavelength ocean flows, because the melting temperature of ice is pressure-dependent (so shell thickness variations cause lateral temperature gradients in the water beneath, e.g. [92]).

It is also expected that horizontal ('zonal') jets will appear. These jets are important because they could potentially transfer momentum to the regions above. Thus, ocean jets might cause non-synchronous rotation (NSR; see below) of the ice shell. Typical ocean circulation velocities depend on the buoyancy flux (i.e. power) available and are expected to be of order cm s^{-1} [93].

Many of these oceans are thought to contain salts, which means that compositional buoyancy is an additional effect that has to be considered alongside the thermal buoyancy arising from bottom heating. Ice shells themselves are expected to be salt-free; thus, an ice shell that melts will deposit fresh (buoyant) water which may not mix effectively with the (denser) salty water below [94]. Thickness changes in the ice shell due to lateral flow of the ice (§2c(ii)) further complicate the picture [88].

Finally, liquid circulation is affected not only by thermal (and compositional) buoyancy, it is also subjected to mechanical forcing, particularly by tides. Tides act on a timescale that is fast compared to the convective timescales. The backwards and forwards 'sloshing' induced by this tidal forcing is not thought to be a significant ocean heat source [95], but it might well be important in generating turbulence that destroys compositional stratification. Oscillating tidal flows may also not average to zero, thereby producing zonal jets [96]. Other mechanical forcings may occasionally have been important; for instance, differential precession might have been the driver for the Moon's early dynamo [97].

3. Surfaces

Planetary surfaces are important because they record the geological history the body has experienced. Cases range from somewhere like Callisto, which has been largely inactive since it formed, to Io, which is so hyperactive that its entire surface is less than 1 Myr old. The three main causes of surface features are impacts, tectonics and (cryo)volcanism, each of which is discussed in more detail below. An excellent summary of our state of knowledge of these processes may be found in *Planetary surfaces* [5].

(a) Impacts

All planetary bodies experience impacts from small bodies orbiting the Sun. These impacts are important not only because for many bodies they produce the dominant surface feature (craters), but also because they provide a way of assessing surface age and geological activity. The classic description of all aspects of impact cratering can be found in [98].

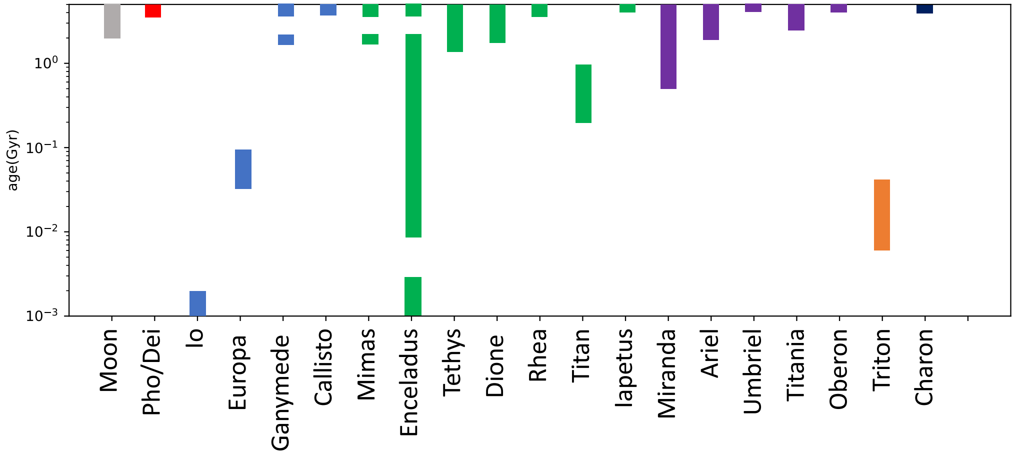

Figure 4. Compilation of approximate surface ages based on crater counts. In most cases, the reference is [104]. Exceptions are: Titan [106], Triton [107] and the Moon [108].

Impacting objects typically strike a surface at tens of km s^{-1}, imparting a large amount of energy. The resulting shock wave causes excavation and melting, and results in a crater which (except for very oblique impacts) is circular. There is a well-understood progression in crater morphology with size, from simple (bowl-shaped) to complex (with an uplifted central peak) to peak-ring or multi-ring basins. This progression can be modified slightly in the case of moons with subsurface oceans. For instance, multi-ring basins such as Valhalla on Callisto (figure 5k) or Tyre and Callanish on Europa may have the form they do because the immediate response to the impact is modified by the finite ice shell thickness. In principle, these basins can be used to place constraints on the ice shell thickness (§2b(iv)). Craters that are anomalously shallow have been observed on some regions of Ganymede, Dione, Tethys, Enceladus and the Moon. These relaxed craters can be used to deduce how high heat fluxes were in the past (§2c(ii)).

For synchronous moons struck primarily by sun-orbiting (heliocentric) debris, the expectation is that the leading hemisphere of the moon will experience more impacts than the trailing hemisphere [99]. Observations show that this expectation is in general not fulfilled. One possible solution to this conundrum is that the moons are experiencing slow non-synchronous rotation (NSR) such that the expected crater distribution is smeared out. NSR may be driven by torques supplied either by tides [100] or by ocean jets [101,102], and is also one possible cause of tectonic deformation on some moons (see §3b). The Saturnian moons are unusual in that some display a population of elliptical craters likely associated with planetocentric, rather than heliocentric, debris [103]. The origin of these impactors is obscure but might be related to a recent catastrophic event in the Saturnian system (§4b).

(i) Surface ages

The surface age of a body is important because it tells us about how active it has been over geological time. Apart from the Moon, where we have samples that can be dated in terrestrial laboratories, the primary way of telling a surface's age is to count the spatial density of impact craters observed. A higher density implies an older surface. Relative ages are therefore easy to determine; absolute ages require a knowledge of the impact flux. In the inner solar system, the lunar record can be used to calibrate this flux, but in the outer solar system the impact fluxes (and corresponding surface ages) are highly uncertain [104]. The picture is further complicated at smaller scales by the presence of 'secondary' craters (i.e. those formed by ejecta from the primary crater). Nonetheless, estimates of surface ages have been made. The key reference is [104]; a more recent compilation is [105].

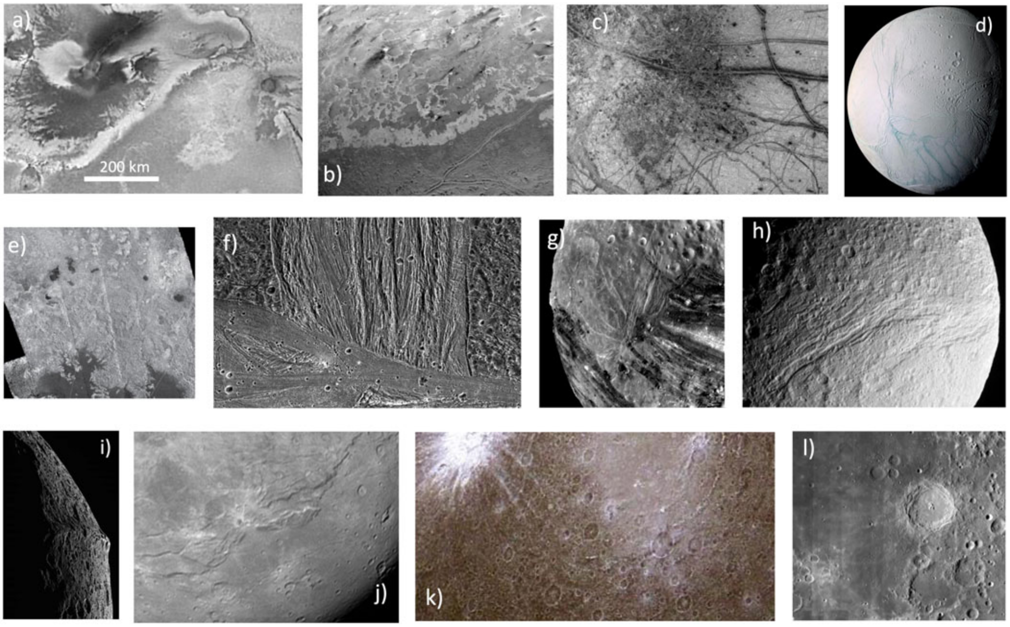

Figure 5. Montage of moon surfaces. Scale is common to all images. Progression is in rough order of geological activity. (a) Io, showing lava flows and channels and individual volcanic vents (NASA planetary photojournal image PIA00537). (b) Triton, showing double ridges and dark deposits from polar plumes (PIA00059). (c) Europa, showing double ridges, dark circular pits, spots and domes, extensional bands and one large impact crater (PIA00294). (d) False-colour view of Enceladus. The geologically active south pole with its four 'tiger stripes' is towards the base of the picture (PIA06254). (e) Radar image of Titan showing channels and dark lakes (mare) including the top of Mare Ligeia (PIA16844). (f) Ganymede, dark terrain cut by north-south grooved terrain, cut by east-west bright terrain (PIA01615). (g) Miranda rifting associated with Arden Corona (PIA00140). (h) The Ithaca Chasma rift on Tethys (PIA09918). (i) Equatorial ridge on Iapetus (PIA08404). (j) Charon, including the Serenity Chasma rift (PIA19932). (k) Heavily cratered Callisto, with one bright impact crater and rings from the Valhalla impact basin (PIA00562). (l) Mare Fecunditatis on the Moon, showing dark mare basalt deposits (left) and heavily cratered terrain (right) (LROC image).

Figure 4 shows a compilation of these ages. While the absolute values should not be taken too seriously, it is evident that moons fall into one of three categories. Some, including Callisto, Rhea, Iapetus, Umbriel, Oberon and Charon, have surface ages in excess of 4 Gyr and have thus experienced little or no geological activity over their histories. Some, including our own Moon, Ganymede, Tethys, Dione, Miranda and Ariel, display some surfaces only 1–3 Gyr old and have thus experienced some activity, perhaps intermittently. Titan is a special case discussed further below. Finally, a handful of satellites show such young surface ages (less than 0.1 Gyr) that they are either known or suspected to be active at present. These unusual objects are Io, Europa, Enceladus and Triton.

Io has no known impact craters on its surface. This is because it is extremely volcanically active, with an average resurfacing rate of about $1\,\mathrm{cm\,yr^{-1}}$. The estimated surface age is about 1 Myr.

Europa also has a young surface, perhaps 50 Myr on average. Here the resurfacing is thought to be primarily tectonic in origin (see §3b). The average resurfacing rate (about $0.6\,\mathrm{km^2\,yr^{-1}}$) is not much smaller than that of the Earth (roughly $3\,\mathrm{km^2\,yr^{-1}}$).

Triton's surface age may be as little as 10 Myr [107]. The cause of the resurfacing is uncertain; it might be due to convective overturn of the ice shell (see §2c(ii)), but there is no consensus on this issue. The ultimate cause of the activity is also uncertain, although obliquity tides seem like the least incredible explanation [45].

Some regions of Enceladus are heavily cratered and therefore old (figure 5d). However, the South Polar Terrain (SPT) is almost free of craters and is perhaps as young as 1 Myr [109],

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owing to the intense tectonic deformation there. Such a wide variation of surface ages on a single body is unusual, although Ganymede has an apparently bimodal age distribution (figure 5f), with the bright terrain perhaps 2 Gyr younger than the ancient dark terrain. How to explain this resurfacing mid-way through Ganymede's evolution is a puzzle (see §4b).

Finally, Titan is an unusual case because of its atmosphere. Small impactors burn up in the atmosphere, so Titan has no small craters. And larger craters are degraded over time because of Titan's active surface processes. The mean surface age is around 0.2–1 Gyr, with strong indications of spatial variations in age [106].

(b) Tectonics

Tectonic features on the surface of a moon are useful because they provide clues to the evolution and dynamics of the body. For instance, a preponderance of compressional features might argue for long-term cooling and contraction (see below). Likewise, some features can be used to infer local heat fluxes, either at the present day or in the past. A good review of planetary tectonic features and their uses may be found in [110].

An important tectonic concept is the lithosphere. Often rather fuzzily defined, it is the near-surface part of a body that is cold and rigid enough not to undergo significant ductile deformation. It is sometimes used as a synonym for the effective elastic thickness, which is the thickness of a purely elastic layer that reproduces the observed deformation behaviour.

Because moons tend to have cold surfaces, near-surface geological materials behave in a brittle fashion when deformed. As a result, tectonic activity typically results in faults and fractures; folding and other forms of ductile deformation like diapirism are comparatively rare. Faults are either normal (indicating extension), reverse (indicating compression) or strike-slip, indicating lateral motion.

The stresses available to drive tectonics can derive from a few different sources. The principal ones are as follows:

(1) Global shape changes. Moons are ellipsoidal (they have a tidal bulge), so if their shape changes or the surface moves relative to the bulge, the lithosphere experiences stresses. Shape changes can arise from change in the moon's spin rate, its tilt, or its total volume (i.e. global contraction or expansion); relative motion of the surface takes place if the surface reorients itself ('true polar wander') or the spin of the moon is not quite synchronous. The stress magnitudes are typically of order MPa.

(2) Local loading. A load, such as a volcanic flow or an impact crater (a negative load), emplaced at the surface will cause stresses in the surrounding lithosphere. Loading may also occur from below, as when a diapir or convective upwelling/downwelling exert stresses on the overlying lithosphere.

(3) Tidal stresses. Moons in eccentric or inclined orbits experience a time-varying gravitational attraction and thus experience time-varying (tidal) stresses at the surface (§2c(i)). The magnitude of these stresses varies with eccentricity and distance to the planet, but are typically significantly smaller than the other stress sources, usually 0.1 MPa or less. Tides are also a source of heat, which can generate stresses directly (through thermal expansion) or indirectly (e.g. via volcanic loading, §3c).

(4) Impacts. As well as producing a long-term negative load (an impact crater), impacts also produce large transient stresses that can fracture the lithosphere. Although they fall off rapidly with distances, the near-field stresses can easily exceed 1 GPa.

As is evident from the montage shown in figure 5, satellites exhibit a very wide range of tectonic activity. One striking takeaway is that on icy bodies exhibiting tectonics, extension is completely dominant. Although compression can be harder to detect, the discrepancy is so large that it is almost certainly telling us something important. Most likely, it means that these bodies have (or had) oceans that re-froze. Freezing involves conversion of water to ice, with the latter taking up more space. On a spherical body, accommodating an increased volume requires outwards motion of the surface, which in turn leads to extension. Rifting on Miranda (figure 5g), Tethys (figure 5h), Charon (figure 5j), Ariel, Dione, Titania and Enceladus, among other bodies, very likely arose from this effect. The dark extensional bands on Europa (figure 5c) may also have arisen in a similar fashion, though Europa is unusual in also possessing apparently compressional features [111]. Iapetus has a unique equatorial ridge (figure 5i), which might be due to compression, but could also be the debris left over from an infalling ring of material [112]. For large icy bodies like Ganymede, the effect of ocean freezing should be less obvious, because high-pressure phases of ice are denser than water and thus counteract the effect of freezing ice I [29]. Nonetheless, Ganymede too is suffused with extensional grooved terrain (figure 5f) with some resurfacing taking place midway through its evolution (§3a(i)).

Although the volume-change effect is probably the dominant source of tectonics on icy bodies, other processes have likely also played a role. Europa is a tectonically complex world (figure 5c); some recent fractures there have been attributed to true polar wander [113], and fluctuating tidal stresses may be the source of the unique curved fractures called cycloids [114]. Local deformation features (pits, spots and domes) are likely the result of intrusion of water or warm ice into the near-surface [115]. At Enceladus, the active tectonics seen at the SPT (figure 5d) appear to be driven by present-day tides [116,117]. The double ridges on Triton (figure 5b) resemble somewhat those on Europa, but are of uncertain origin. One possibility is that both features are the result of tidally-driven frictional heating along the ridge [118]. Triton also possesses unique 'cantaloupe terrain'. This is thought to have arisen from deformation due to rising diapirs, similar to the salt diapirs seen on the Earth [75].

Impacts typically cause only local deformation features; however, there are multi-ring structures on both Ganymede and Callisto (figure 5k) that extend for thousands of km and are due to the transient stresses imposed by large impacts. The symmetric orientation of coronae on Miranda is almost certainly because of true polar wander [76], but the original source of the stresses causing these features is not certain (one favoured explanation is convective upwellings [119]). Enceladus likewise shows a globally symmetric tectonic pattern, with the regions of greatest deformation being the south pole and the centers of the leading and trailing hemispheres [120]. This geometry may also be due to an episode of true polar wander.

Silicate bodies (Io and the Moon) are different in an important respect from icy moons: they display compressional features. On the Moon these take the form of reverse faults (lobate scarps and wrinkle ridges), while on Io they are typically uplifted crustal blocks (mountains). For the Moon, the primary cause of this compression is the slow cooling and contraction of the entire body over billions of years. Interestingly, the Moon is not completely dormant at present: the *Apollo* mission recorded moonquakes, some of which are triggered by tides; and there are recent (less than 100 Myr) tectonic features observed, at least some of which are extensional [121].

For Io, compressional features arise because volcanic activity rapidly buries surface layers; as these layers are pushed deeper, they have to contract (because Io is a sphere) and as a result large (hundreds of MPa) compressive stresses are generated [122].

The small Martian moon Phobos displays an unusual, global set of grooves [123]. Although some of these may be the result of chains of impacts ('catenae'), others appear to be tectonic in origin. The most likely explanation is that these features reflect that Phobos is drifting inwards towards Mars and is getting slowly ripped apart by increasing tidal stresses [124]. Some small moons in the Saturnian system display somewhat similar features [125].

(c) Volcanism and cryovolcanism

Some bodies are sufficiently hot in the interior that they produce melt, either magma (for silicates) or water (for icy bodies). Volcanism involves the upwards motion of melt; eruptions at the surface are termed effusive volcanism, while emplacement of melt in the subsurface is termed intrusive. Water-based volcanism on icy bodies is termed cryovolcanism. These processes are important because they can produce surface features which tell us about conditions and dynamics in the subsurface. They also potentially represent an important method of heat transport. The relevant chapter in [5] has a good summary; a qualitative survey may be found in [126].

The most important difference between volcanism and cryovolcanism is that, typically, magma is less dense than its host rock, and is thus positively buoyant, whereas water is denser than ice. As a result, silicate melts have a natural tendency to rise towards the surface, whereas water is more likely to be stranded at depth. There are ways of overcoming this negative buoyancy; for instance, if gases in the water start to come out of solution (as in a shaken soda bottle) [127] or the liquid is ammonia-rich [128] then the negative buoyancy is reduced. Alternatively, ocean pressurization due to shell freezing may force the water upwards [28]. Nonetheless, there is a general expectation that cryovolcanism is less likely to occur than silicate volcanism.

Because volcanism involves melting, it requires an energy source (which is often tidal heating). Likewise, the eruption of warm material to the cold surface involves an advective transfer of heat which can in some cases dwarf the conduction of heat through the lithosphere [77]. For explosive eruptions, such as Io's volcanic plumes, the velocity of erupting material can be used to determine the entropy content of the source [129].

The morphology of lava flows depends primarily on their viscosity. This depends on the eruption temperature and also (for silicates) the concentration of Si. Water has a very low viscosity (approx. 10^{-3} Pa s) and is thus expected to form thin, smooth flows, unless it has a very high ice crystal content.

(i) Examples of silicate volcanism

Io, famously, is the most volcanically active body in the solar system. Its average resurfacing rate of $1\,\mathrm{cm\,yr^{-1}}$ results in an advective heat flux of about $2.5\,\mathrm{W\,m^{-2}}$ [70]. Some volcanic plumes jet material 400 km into space, implying exit velocities of over $1\,\mathrm{km\,s^{-1}}$. Some of the lavas erupted have temperatures as high as the hottest lavas on the Earth, exceeding 1700 K [130]. Io's extreme volcanism causes surface burial and compression (§3b) and is a consequence of the tidal heating it experiences.

The Moon is not volcanically active now, but shows extensive evidence of ancient volcanism [108]. The extensive dark, smooth plains (maria) (figure 5l) are the product of ancient lava flows. This volcanism, which declined over at least 2 Gyr, is a consequence of the ancient Moon starting life molten and cooling gradually with time.

(ii) Examples of active cryovolcanism

There are only two bodies exhibiting confirmed cryovolcanic activity at the present day: Enceladus and Triton. It has been argued that Europa might have such activity, but none of the current evidence is especially convincing.

Enceladus is the poster child for cryovolcanism. One of the biggest surprises of the Cassini mission was the detection of water vapour and ice grains being ejected from the SPT of Enceladus, with vapour speeds of perhaps $800\,\mathrm{m\,s^{-1}}$ [131]. Because some of the grains are salty, the source must be a salty ocean beneath the ice [17]. The eruptions occur along the four polar 'tiger stripes' (figure 5d), although whether the sources are points or extended is uncertain. The presumption is that the ocean is connected to the surface via narrow, water-filled cracks. The eruptions are modulated over the course of an orbit, indicating that tidal stresses are causing the crack widths to vary [116]. Perhaps the most important consequence of Enceladus's behaviour is that it provides a way of obtaining samples of ocean material without having to drill through kilometres of ice. As a result, Enceladus is a high priority for future spacecraft missions [6].

Triton also has eruptive polar plumes (figure 5b), although these are only 8 km or so high and are of unknown composition. The mechanism driving the eruption is also unknown—the plumes could be similar to those at Enceladus, but they might also be driven by near-surface solar heating [132].

Earth-based telescopic observations [133,134] and in situ measurements [135] have been used to argue for the presence of eruptive plumes on Europa. However, the results are generally either low signal-to-noise or non-unique, and the claimed plumes are so large that the exit velocities implied are hard to understand on theoretical grounds. Furthermore, the plumes, if they exist, are intermittent—many observations (including a recent JWST campaign [136]) show no evidence of plumes.

(iii) Ancient cryovolcanism

At some point, almost every icy satellite has been argued to show evidence for cryovolcanism. However, in several cases subsequent higher-resolution images have shown that supposed cryovolcanic deposits are actually tectonic in origin [137], so scepticism is warranted.

Europa is one place where cryovolcanism seems likely, although present-day activity is not confirmed. Some linear double ridges may have erupted the reddish material observed on their flanks [138]. Near-surface melt may have been intruded beneath the so-called chaotic terrain, and intrusions are likely the cause of the pits, spots and domes seen on the surface [115] (figure 5c).

Smooth plains on Charon (figure 5j), smooth bright terrain on Ganymede (figure 5f), and smooth or lobate deposits on Ariel, Dione and Triton have all been proposed as cryovolcanic deposits. Morphology alone, however, does not provide a very convincing argument. This is particularly true for Titan, where imaging resolution is poor. Despite much work, there are currently no features on Titan that are generally accepted as cryovolcanic [137].

(d) Other processes

The surface appearances of most moons are controlled by some combination of impacts, tectonics and (perhaps) cryovolcanism. Because most of these bodies lack atmospheres, wind processes only happen where the local atmospheric pressure increases transiently, e.g. due to erupted gases. Thus, Io exhibits a few dunes and Triton shows dust deposits associated with its plumes.

The one major exception is Titan, which has a thick (1.5 bar) atmosphere and a ‘hydrological cycle’ in which methane is stored in surface lakes and seas, evaporates and is eventually rained out from the atmosphere [139]. Uniquely among moons, therefore, Titan has rivers and channels, evidence of surface flow (figure 5e). The margins of its lakes and seas may be modified by wave action, and there may be dissolution features reminiscent of limestone (karst) terrains on the Earth. Titan also has fields of dark longitudinal dunes, which are shaped by the prevailing winds and probably consist of small organic grains [140].

Another minor process is mass-wasting, i.e. landslides. Areas with steep topography sometimes show mass-wasting features, such as crater walls on Callisto and steep-sided mountains on Io. Sublimation and redeposition of volatile materials can also occur, modulated by seasonal insolation variations. Examples include the extreme albedo variations on Iapetus [141], the reddish polar cap on Charon [142], and redistribution of volatile sulfur compounds across the surface of Io.

Finally, surface properties can also be affected by exogenic processes such as energetic particle or micrometeorite bombardment (e.g. [16]). These processes can alter the colour, crystallinity and composition of near-surface materials, often on timescales that are short compared with other geological processes. A good example is the spatial variations in colour observed across the Saturnian satellites, a consequence of both plasma and dust bombardment [143]. Another is the Moon, where local albedo features (‘swirls’) arise because of the deflection of charged particles by strong local magnetic fields [144].

4. Formation and long-term evolution

The narrative above has mainly focused on describing the present-day properties of the moons. However, it is also of interest to understand how they came to have those present-day properties, and in particular what records (if any) remain of how they formed and evolved. These questions are more difficult to assess, and will be discussed fairly briefly in what follows. But they are important; for instance, satellite formation cannot be cleanly separated from how the host planet formed, and the details of planet formation are of great interest both for this solar system and for the thousands of exoplanetary systems discovered. A dated but still useful review may be found in [145].

Some of the satellites appear to have surface ages similar to that of the solar system (§3a(i)). As a consequence, they may have preserved evidence of the processes by which they originally formed; one such body is Callisto (see below). Somewhat more active satellites may have lost this early information, but instead preserved a record of the long-term evolution of the body. The diminishing volcanic activity on the Moon over geological time (§3c(i)) is one such example; the resurfacing of Ganymede is another.

(a) Satellite formation

Most satellites formed primarily within a disk of material accreting onto their parent planet, although heliocentric debris may also have contributed (and our own Moon is a special case). The local thermal environment may have been influenced by the parent planet's evolving luminosity, as well as insolation and viscous dissipation in the proto-satellite disk. Satellite formation models are well summarized by Canup & Ward [146] and McKinnon [147]. At present, the details of how, and how fast, these moons formed are poorly known.

Determining the process of satellite formation from observations is difficult. There are three primary observations of interest. The first is compositional data such as the volatile (e.g. ice) content or isotopic ratios. The former might tell us something about the solid material available in the proto-satellite disk, and thus potentially its thermal structure (e.g. [148,149]); the gradient of density in the Galilean satellites (§2a) is a case in point. The latter might tell us about provenance or origins (e.g. did Phobos originate from Mars or not? Why is our Moon depleted in volatile elements?).

The second is the orbital characteristics of these bodies. For instance, the fact that Triton is retrograde strongly suggests that it was captured and originally resided in the Kuiper Belt [67]. Likewise, the fact that Phobos is migrating inwards places constraints on where it could have originated from [150].

The third concerns the extent to which the body is differentiated, as measured by its MoI (§2b(ii)). For a large icy satellite which accretes relatively rapidly, the gravitational energy released during accretion (§2b(ii)) is enough to cause melting and differentiation. Conversely, if the body is *not* fully differentiated, that places tight constraints on how rapidly it could have been put together. The key example is Callisto. *If* Callisto is not fully differentiated (which rests on the undemonstrated assumption of hydrostatic equilibrium; see §2b(ii)), then it must have accreted on a timescale longer than 10^5 years [148]. This timescale in turn places constraints on the density and dynamics of the proto-satellite disk.

Our own Moon provides another example, as reviewed in [151]. The Moon's MoI (§2b(ii)) implies a small iron core. This observation is readily explained if a giant impact onto the Earth spalled off strongly heated material that eventually re-accreted to form the Moon. Since the bulk of the spalled material will have come from heated portions of the Earth's mantle, the small core of the Moon—and its depletion in volatile elements—is the expected outcome. The extreme isotopic similarity between the Earth and Moon have complicated this simple picture [152], but not overturned the general concept of a Moon-forming impact.

(b) Satellite evolution

The key aspect of satellite evolution that makes it different from planetary evolution is that the main energy source driving satellite evolution is tidal heating. This means that the long-term thermal and orbital evolution of these bodies is intimately connected, and complex; orbital evolution depends on the state of the satellite interior via k_2/Q (equation (2.1)), but the state of the interior will be affected by orbital characteristics via tidal heating. As a result of these feedbacks, satellites can experience periodic heating episodes and non-monotonic temperature evolution (e.g. [1]). Although there does not appear to be a good general review of these processes, Nimmo *et al*. [153] provides an example of how they apply at Enceladus.

(i) Orbital and thermal evolution

Why does orbital evolution happen? The answer is that satellites raise a bulge on the primary. If the primary's response lags by an angle $1/Q_p$, where Q_p is its dissipation factor, then the primary exerts a net torque on the satellite and pushes the satellite outwards, while its own spin period lengthens. This process is why the Moon is still moving away from the Earth at $4\,\mathrm{cm\,yr^{-1}}$. Recently, astrometric measurements of satellite migration have been used to determine the Q_p values of Jupiter, Saturn and Uranus [65,154,155]. For the former, $Q_p \sim 10^5$, indicating that Jupiter is not very dissipative, but for Saturn and Uranus the values are smaller ($Q_p < 10^3$). The physical explanation for this behaviour may be due to the so-called 'resonance locking' in which the satellite migration timescale is tied to the evolution timescale of the planet's interior [156]. A low effective Q_p means that the Saturnian and Uranian moons may have migrated further than traditionally thought, opening up their evolutionary parameter space. Because Q_p also controls how rapidly energy can be transferred from a planet to its moons [157], a low Q_p also helps to explain how Enceladus's high heat output is sustained [153,156] and the high paleo-heat fluxes inferred for the Uranian moons (e.g. [57]).

In general, closer-in satellites are expected to migrate outwards more rapidly. They can thus catch up to their more slowly moving outer companions and potentially get trapped into some kind of orbital resonance with them.[3] However, not all resonances are stable—potential resonances between the Uranian satellites overlap, leading to chaotic behaviour [158]—and some resonances can break spontaneously if the moons' eccentricities get too large. Three-body and higher-order resonances complicate the picture still further (e.g. [159]).

Because of this complexity, little can be said definitively about the orbital evolution of most satellites. The main exception is the Moon, which is (relatively speaking) a simple and well-characterized system. However, one important recent discovery is that the Earth–Moon angular momentum budget is not a fixed quantity, as was previously thought, but may have been larger in the past before being reduced as the Moon went through the so-called evection resonance with the Sun [160]. The Moon's ability to generate a magnetic field early in its history may also be linked to its orbital evolution [97], though other explanations for its ancient dynamo also exist.

There are other two-body systems that are relatively simple. The Pluto-Charon system is doubly synchronous (the end-state of tidal evolution); Charon may have experienced early tidal heating if its orbit was initially non-circular [161]. Phobos is spiralling inwards towards Mars, because it is orbiting faster than Mars spins; it will be torn apart in another 50 Myr. Triton is also slowly spiralling inwards, because is orbiting retograde. Its eccentricity is almost zero, as expected for an isolated satellite (§2c(i)), but its obliquity may be quite large [45].

Systems with multiple satellites are harder to understand. The three inner Galilean satellites are in a 1 : 2 : 4 resonance (the Laplace resonance) which keeps Io and Europa's eccentricities excited. This system appears to be close to equilibrium, whereby the eccentricity-damping effect of tidal heating in Io is balanced by the eccentricity-increasing properties of the Laplace resonance [65]. How and when the Laplace resonance originated is an open question. One clue may be provided by Ganymede, which apparently experienced a pulse of tidal heating around 2 billion years ago (see §3a). Two possible explanations are that prior to being captured into the present-day resonance, Ganymede either went through a 'Laplace-like' resonance [29] or experienced chaotic interactions with Europa [162].

Things are even more complicated in the Saturn system. There are two pairs of overlapping resonances, an eccentricity-type resonance between Enceladus and Dione, and an inclination-type resonance between Mimas and Tethys, while Titan is in a 4 : 3 resonance with Hyperion. These resonances presumably arose as the satellites were pushed out by tidal torques from Saturn, but

[3]Note that resonance-trapping only occurs on *converging* orbits; see [63].

the details are unclear [163,164]. Evidence for past heating events on some of the Saturnian moons (§2c(ii)) also suggests a complex history [164]. Furthermore, there are several hints (e.g. high eccentricity of Titan, the apparently young age of Saturn's rings, the unusual cratering record) that the Saturn system may have undergone some relatively recent catastrophic event (e.g. [69]). If so, then much of the evidence for earlier epochs of evolution may have been lost.

Finally, none of the Uranian satellites are in resonances at the present day, so their eccentricities are correspondingly small. However, geophysical observations (e.g. [57]) suggest past episodes of high heat flux which require high eccentricities presumably arising from past resonances and a low Q for Uranus, consistent with astrometric measurements [155]. Models of Uranian satellite evolution are hampered by the very limited data available, but are an area of active research [68].

(ii) Compositional and surface evolution

Whether any satellites underwent significant post-formation evolution in their bulk composition is an open question. One possibility is that the gradient in composition observed in the Galilean satellite system (figure 2) is a consequence of long-term evolution rather than conditions of formation. Since tidal heating falls off rapidly with semi-major axis, it has been proposed that Io and Europa lost all and most of their volatiles, respectively, due to long-term tidal heating [146]. However, detailed investigations do not support this hypothesis [165].

The disorganized pattern of densities observed in the Saturnian system (figure 2) is also unexplained. The very low density of Tethys suggests it might be an icy fragment resulting from an impact onto a much larger, now-vanished moon [166]. Titan's eccentricity may likewise have been excited by a recent (approx. 100 Myr) encounter with a smaller body that was eventually tidally disrupted to form Saturn's dramatic rings [69]. Although the evidence for it is equivocal, any late scattering of material inwards from the Kuiper Belt [167] could potentially have broken apart many of the small, inner moons, e.g. [168], likely causing changes in bulk composition.

The larger moons may have undergone limited post-accretion compositional evolution driven by radioactive heating. For instance, Europa's (putative) iron core may not have formed for a billion years or so [30], while its ocean could in principle have been 'sweated out' of hydrated minerals, rather than being a primordial feature [169]. Absent tidal heating, existing liquid layers will have started to re-freeze as radiogenic heating decayed. This gradual refreezing will have increased the concentration of incompatible elements (e.g. sulfur in cores, NH_3 in oceans) in the remaining liquid, potentially lengthening the liquid layer's life-time because of their anti-freeze effects (§2a, e.g. [170]).

As discussed in §3, most of our information about satellite evolution comes from observations of surface features. Thus, episodes of ancient tidal heating at Ganymede and the Uranian and Saturnian satellites are inferred from tectonic surface features. The preponderance of extensional tectonics on the mid-sized satellites can naturally be explained by refreezing of subsurface oceans, while the Moon's dominantly compressional tectonics is a consequence of its slow cooling and contraction. All bodies will have accumulated impact craters and developed surface regoliths. In many cases, including the Moon, energetic particles will cause long-term changes in the optical and chemical properties of the near-surface. Volatile species will evaporate, migrate and re-condense in response to seasonal or longer-term variations in insolation.

Some satellites may have experienced long-term volatile loss. Titan's atmosphere has experienced some N loss over billion-year timescales; the fact that it still contains CH_4 (which has a short lifetime) suggests replenishment from the interior [171]. How Titan's atmosphere originated, however, and what explains the absence of an atmosphere on similarly massive Ganymede, is an open question. Enceladus is losing ice to outer space at a current rate of order $10\,\mathrm{kg\,s^{-1}}$, enough to change its bulk composition by a few per cent if sustained over 4 Gyr. The extreme isotopic fractionation of S and Cl measured at Io imply that its rate of loss from the exosphere and volcansic outgassing have been maintained at roughly their present levels over the age of the solar system [19].

5. Summary and conclusion

Over the last 30 years, the moons of the outer solar system have been explored by four spacecraft (*Galileo, Cassini*, *New Horizons* and *Juno*), and our understanding of these bodies has advanced correspondingly. More recently, lunar studies have seen a renaissance thanks to orbital (such as *GRAIL*, Korean Pathfinder Lunar Orbiter and *Chang'e 1* and *2*), lander and sample return missions (*Chang'e 3-6*). Phobos and Deimos were examined by *Mars Express* and a Phobos sample return mission (*MMX*) is planned. What are the biggest intellectual advances arising from these studies?

Perhaps the single biggest surprise was the present-day activity at Enceladus discovered by *Cassini*. The fact that this tiny moon was venting material into space was unexpected, and illustrates the supreme importance of tidal heating (as well as our inability to make predictions). Largely because of its provision of accessible samples of a subsurface ocean, Enceladus is now a very high priority for future spacecraft exploration [6].

One general advance is the recognition of the abundance of ocean worlds. Thirty years ago, it was not clear that any of the bodies depicted in figure 1 harboured subsurface oceans. Now, we are fairly certain that at least five do, and suspect that others do as well. This recognition has implications for the potential habitability of both this and other planetary systems.

Another is the recognition that orbital evolution can be more rapid than previously thought, and also frequency-dependent. This result means the energy available to drive tidal heating in satellites may be greater than previously thought, and the parameter space of possible orbital evolution scenarios is wider. In this context, a better theoretical understanding of gas giant dissipation (e.g. [172]) would be highly desirable.

Finally, three decades ago the origin of the Moon was thought to be a well-understood problem. Since then, the recognition that its angular momentum budget may not have been conserved, and the almost perfect isotopic match between Moon and the Earth have challenged conventional assumptions about the Moon's formation. The current picture is confused and as yet there is no consensus regarding the most likely answer.

The formation of the Moon thus represents a major puzzle at present. Equally, neither the origin of Phobos and Deimos, or the small satellites of Pluto [173], have agreed-upon solutions. What are some other major puzzles?

One long-standing puzzle is Titan's atmosphere. Why should it have a thick atmosphere, while Ganymede and Callisto (almost identical in mass) do not? Perhaps this is simply an outcome of stochastic impact processes [174], or a recent upheaval in the Saturn system [69], but the answer is not currently clear.

The origin of Ganymede's partial resurfacing, and more generally the origin and evolution of the Laplace resonance, also remain open questions. These issues are complicated by the potentially low and/or frequency-dependent Q_p of Jupiter, which has so far been little explored (though see [175]). A possibly related, and equally long-standing, puzzle is whether Callisto is really differentiated or not.

Answering these and other puzzles will in most cases require future spacecraft missions. For instance, compositional measurements (and especially sample return) are likely to be of fundamental importance. Thus, whether Phobos is a fragment of Mars or not should be determined once samples are delivered to the Earth by *MMX*. Similarly, *in situ* measurements made by the forthcoming *Dragonfly* spacecraft should clarify the origin of Titan's atmosphere. Sample return from previously unsampled regions of the Moon by *Chang'e 6* and potentially other spacecraft may help pin down its formation and early evolution.

Not all compositional measurements require spacecraft. For instance, isotopic ratios at both Io [19] and Europa [136] have been measured using remote sensing from the Earth. In the case of Io, these measurements provide the first evidence that tidal heating there has operated for billions of years and is not just a recent phenomenon. Ground-based capabilities such as these are likely to be of increasing importance in the next decade, not least because of a dearth of forthcoming spacecraft observations in the outer solar system. They may also finally find the first exo-moons [176].

The dynamics of subsurface oceans (§2c(iii)) is a very rich theoretical field. The difficulty is in obtaining *observations* that can be used to discriminate between the various models proposed. Currently, the best sources of information are the spatial distribution of surface features and shell thicknesses. High precision rotation state data, either from spacecraft or from the Earth [177], are likely to be very useful in this context. Magnetic induction could in principle be used to determine ocean flow velocities, but the signals are likely to be too small to be useful.

Another rich theoretical field, albeit one with a much longer history, is orbital dynamics. Recent studies have clarified the importance of three-body orbital interactions—as illustrated by the Moon, and the Uranian and Saturnian satellites. The potentially strongly frequency-dependent dissipation inside the parent planets also needs to be accounted for. And essentially all orbital evolution scenarios should take into account thermal–orbital feedbacks. Taken together, these newer effects have so expanded the parameter space that exploring it is a daunting task.

After about 2032, a flotilla of spacecraft will start accumulating data. Both *Europa Clipper* and JUICE will commence their investigations of Europa and Ganymede, respectively; the latter will provide geodetic data of a quality previously only available for the Earth and Moon. *JUICE*, perhaps aided by *Tianwen-4*, will also lay to rest the question of whether Callisto is only partially diffferentiated. *Dragonfly* is expected to start operations at Titan in 2034. Lunar exploration will probably continue at an increasing (and increasingly multi-national) rate, perhaps even with a crewed component. Looking even further ahead, Enceladus is a high-priority target for both NASA and ESA, while the next US Flagship mission will (eventually) explore the Uranus system [6].

This article has attempted to summarize how much our understanding of the surfaces, interiors and evolution of solar system moons has matured over the last 30 years. But there is a great deal that still remains either unexplained or unexplored. Although some patience will be required, the next flotilla of spacecraft should provide new generations of planetary scientists answers to existing puzzles, and generate many new ones.

Data accessibility. This article does not contain any additional data.

Declaration of AI use. I have not used AI-assisted technologies in creating this article.

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